

BRAIN-COMPUTER INTERFACE

Brain-computer interfacing will make it possible to get a task done by just thinking about it. Take a look at the system, research areas and possible applications

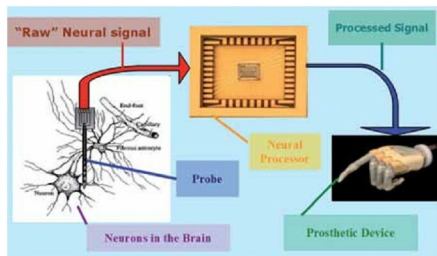
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Just imagine you have returned home after a long day's work in a frosty night. As you near the main door of your house, the door opens up sensing your arrival. Your wish to drink a cappuccino commands the coffee maker to prepare that perfect coffee. The foot massager knows just how hard to massage your feet and calves.

This might be possible with brain-computer interfacing in the near future. Brain-computer interface (BCI) provides a means for communication between the human brain and a computer, so the information of neural activities in the brain can be exchanged with the computer. The computer can, in turn, use the information to control a variety of devices or exchange the information with another BCI. Thus BCI extracts electrophysiological signals from suitable components of the brain (motor cortices) and processes them to generate control signals for computers, robotic machines or communication devices.

BCI is a highly interdisciplinary field involving neurology, mathematics, electronics engineering, material science, etc. Advancements in these fields have led to astonishing accomplishments in brain-computer interfacing. Imagine how exciting it would be to get rid of the handheld devices and be able to communicate through mind.

Ongoing research on transferring the mind to a computer is the ultimate endeavour in the entire history of humanity that will take human evolution to a new level altogether. If you take a quick look into the possible applications of BCI, you will see the oblation of handheld devices and communication becoming seamless. With



Building blocks of a typical BCI system

intelligent (and emotional) devices all around us, it will become possible to just think about a task to be done and it will be done!

Now think about this situation: A brain-computer-interfaced object accessed by a number of people to serve their purpose. The object gets confused since so many people have wished it to serve their purpose at the same time. Thus such devices will have to have intelligence to resolve the deadlock of multiple requests and do the prioritisation to serve a particular request.

Building blocks of a BCI system

Probes are implanted in the brain to tap neural signals. Depending upon the type of signal to be tapped and analysed, probe location is decided. Ideally, the probe would incorporate a signal booster circuit to strengthen the weak neural signals and transfer them to a neural processor. The neural processor works upon raw neural signals to extract the different information. For example, if the signal was meant to

control a finger-movement, the neural processor would decipher this and generate a control signal to apply appropriate stimuli to the prosthetic arm.

The important components of a BCI system are explained below.

Neurons in the brain. Neurons are signal carriers in the brain that transfer information from sensory inputs of the body (eyes, ears, tongue, nose and skin) to the processing unit of the brain (hippocampus).

Challenging areas in understanding the neurons are:

1. **Neural signal behaviour.** There are over 100 billion neurons in the brain. Identifying them and understanding their behaviour is a daunting task, much like identifying and understanding stars in the universe. Electromagnetic signals of such closely packed neurons induce noise and it becomes difficult to associate a signal with its actual neural-source. This makes the study of neural-signal behaviour very challenging.

Some of the techniques to read-out the brain are electroencephalography